

# Lecture 2: From Lattices to Learning With Errors

SVP, CVP, SIS, LWE, Ring-LWE, and Module-LWE

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Lecture Notes — Session 2 (approx. 2 hours)

**Purpose of this lecture.** Lecture 1 built the algebraic toolkit — *Group*  $\rightarrow$  *Subgroup*  $\rightarrow$  *Ring*  $\rightarrow$  *Field*  $\rightarrow$  *Modular Arithmetic*  $\rightarrow$  *Vector Space*  $\rightarrow$  *Basis*  $\rightarrow$  *Lattice*  $\rightarrow$  *Determinant* — and ended by naming the Shortest Vector Problem (SVP) as one of the hard problems lattice cryptography rests on. This lecture finishes the job: we state SVP and CVP formally, meet the **Short Integer Solution (SIS)** and **Learning With Errors (LWE)** problems precisely, and then generalize LWE to **Ring-LWE** and **Module-LWE** — the exact hard problems underneath Kyber (ML-KEM) and Dilithium (ML-DSA). By the end, the public key of a real post-quantum scheme should look to you like “just an LWE sample,” not a mysterious block of numbers.

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## 1 Quick Recap of Lecture 1

Before building anything new, let’s fix the vocabulary we’ll be using constantly today.

| Term                     | One-line meaning                                                               |
|--------------------------|--------------------------------------------------------------------------------|
| Ring $(R, +, \times)$    | Two operations; addition invertible, multiplication need not be                |
| $\mathbb{Z}_q$           | Integers $\{0, \dots, q-1\}$ , wraparound (“clock”) arithmetic                 |
| Vector space, basis      | Every point = a unique combination of basis directions                         |
| Lattice $L$              | <i>Integer-only</i> combinations of a basis; a discrete grid in $\mathbb{R}^n$ |
| $\det(L)$                | Volume of the lattice’s fundamental cell; basis-independent                    |
| $\lambda_1(L)$           | Length of the shortest nonzero vector in $L$                                   |
| Good basis vs. bad basis | Short, near-orthogonal vs. long, skewed — same lattice, different difficulty   |

### Remark: The One Idea Everything Today Builds On

Lecture 1 ended with this sentence: “*the secret key is typically a good (short) basis, while the public key describes the same lattice using a bad basis.*” Today we make that idea concrete and computational. Every hard problem in this lecture is, underneath, a question of the form: “*given a bad description of a lattice, recover something a good description would make trivial.*”

## 2 Two Canonical Hard Problems on a Lattice

### 2.1 The Shortest Vector Problem (SVP)

#### Definition: Shortest Vector Problem (SVP)

Given a basis  $B = \{b_1, \dots, b_n\}$  for a lattice  $L = \mathcal{L}(B) \subseteq \mathbb{R}^n$ , find a nonzero vector  $v \in L$  with  $\|v\| = \lambda_1(L)$  (the minimum distance defined in Lecture 1). The **decision** version,  $\text{GapSVP}_\gamma$ , only asks: is  $\lambda_1(L) \leq d$ , or is  $\lambda_1(L) > \gamma \cdot d$ , for a given distance  $d$  and approximation factor  $\gamma \geq 1$ ? (One of these two must hold; the problem promises no “messy middle” case.)

#### Remark: Why the “Approximation Factor” $\gamma$ Matters

Finding the *exact* shortest vector ( $\gamma = 1$ ) is extremely hard even for classical computers once  $n$  is a few hundred. But so is finding a vector even *polynomially longer* than the shortest one, as long as  $\gamma$  stays modest and  $n$  is large — and that weaker, “approximate” version is all cryptography actually needs to be hard. This is why you will see phrases like “SVP is hard to approximate within polynomial factors” rather than just “SVP is hard.”

### 2.2 The Closest Vector Problem (CVP)

#### Definition: Closest Vector Problem (CVP)

Given a basis  $B$  for a lattice  $L \subseteq \mathbb{R}^n$  and a **target point**  $t \in \mathbb{R}^n$  (not necessarily in  $L$ ), find the lattice point  $v \in L$  minimizing  $\|t - v\|$ . The approximate decision version  $\text{GapCVP}_\gamma$  again just distinguishes “some lattice point within distance  $d$  of  $t$ ” from “every lattice point is farther than  $\gamma \cdot d$  from  $t$ .”

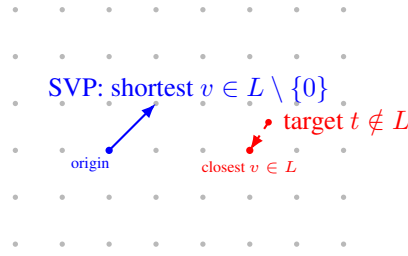


Figure 1: SVP (left, blue) asks for the shortest nonzero vector reachable inside the lattice itself. CVP (right, red) asks, for an arbitrary target point that is *not* on the grid, which grid point is nearest. Both are easy to state and easy to solve by eye in 2D — and famously hard in high dimensions with only a bad basis to work with.

#### Example: CVP by Eye vs. CVP by Formula

In Figure 1 you can *see* the closest grid point instantly. That’s because your eye is effectively using a very good, orthogonal basis (the obvious horizontal/vertical grid directions). Algorithmically, CVP is solved by writing  $t = \sum c_i b_i$  (allowing real, non-integer  $c_i$ ) and **rounding each  $c_i$  to the nearest integer** — but this shortcut only gives the *true* closest point when the basis is good (short, near-orthogonal). Hand the exact same lattice to an algorithm with a bad basis instead, and naive rounding can miss the true closest point by a wide margin. This gap — correct-with-a-good-basis, unreliable-with-a-bad-basis — is precisely what CVP-based cryptography exploits.

#### Looking Ahead: Where SVP and CVP Are Heading

We will not attack SVP or CVP directly today. Instead, both SIS (Section 3) and LWE (Section 4) are best understood as *specific, structured* instances of SVP/CVP, dressed up so that a random instance is easy to *generate* but — conjecturally — just as hard to *solve* as the worst case. That “random instances are as hard as the worst case” guarantee is the single most important design property of modern lattice cryptography, and we return to it in Section 4.4.

### 3 The Short Integer Solution (SIS) Problem

#### Remark: Motivating SIS With a Question

Here’s a question that sounds almost too easy: if I hand you a random matrix  $A$  (think: rows and columns of random numbers mod  $q$ ), can you find a *short* nonzero vector  $x$  so that  $Ax \equiv \vec{0} \pmod{q}$ ? If  $x$  could be any size at all, this would be trivial linear algebra — as long as  $A$  has more columns than rows, there are always *some* nonzero solutions (more unknowns than equations). The catch, and the entire reason this is a cryptographic problem rather than a homework exercise, is the word **short**: among the (many) solutions that exist, finding one whose entries are all small turns out to be hard once the matrix is big enough. This is SIS.

#### Definition: Short Integer Solution (SIS)

Fix positive integers  $n, m > n$ , a modulus  $q$ , and a bound  $\beta$ . Given a uniformly random matrix  $A \in \mathbb{Z}_q^{n \times m}$ , find a nonzero vector  $x \in \mathbb{Z}^m$  with

$$Ax \equiv \vec{0} \pmod{q} \quad \text{and} \quad \|x\| \leq \beta.$$

### Remark: Why This Is Secretly a Lattice Problem

Define  $\Lambda^\perp(A) = \{x \in \mathbb{Z}^m : Ax \equiv \vec{0} \pmod{q}\}$ . A quick check confirms  $\Lambda^\perp(A)$  is a subgroup of  $\mathbb{Z}^m$  (closed under addition and negation, contains  $\vec{0}$ ) and it is discrete (it's a subset of  $\mathbb{Z}^m$ ) — so by Lecture 1's subgroup definition,  $\Lambda^\perp(A)$  **is itself a lattice**. Solving SIS is exactly finding a short nonzero vector in  $\Lambda^\perp(A)$  — i.e., **SIS is SVP**, restricted to this specific, algebraically-generated family of lattices. The reason cryptographers like this family is that generating a *random* instance is as easy as generating a random matrix  $A$ , yet the resulting lattice is (conjecturally) exactly as hard to solve SVP on as the worst-case lattices Lecture 1 discussed abstractly.

### Example: A Tiny SIS Instance by Hand

Let  $n = 1$ ,  $m = 4$ ,  $q = 17$ , and  $A = \begin{pmatrix} 3 & 5 & 8 & 1 \end{pmatrix} \in \mathbb{Z}_{17}^{1 \times 4}$ . We want a short, nonzero, *integer* vector  $x = (x_1, x_2, x_3, x_4)$  with  $3x_1 + 5x_2 + 8x_3 + x_4 \equiv 0 \pmod{17}$ . Try  $x = (1, 1, -1, -16) \rightarrow$  too long. Instead try  $x = (1, -1, 0, 2)$ :  $3(1) + 5(-1) + 8(0) + 1(2) = 3 - 5 + 0 + 2 = 0 \equiv 0 \pmod{17}$ . Success, and  $\|x\|$  is small (each entry is  $\leq 1$  in absolute value except the last, which is 2). This  $x$  is a valid, short solution — with  $m = 4$  columns squeezed against only  $n = 1$  modular equation, short solutions are guaranteed to exist by a pigeonhole argument (more columns than rows means lots of redundancy); the point of SIS being *hard* is that finding one is nontrivial once  $n, m$  are cryptographically large (hundreds to thousands), not that one fails to exist.

### Remark: What SIS Is Used For

SIS is the workhorse behind lattice-based **hash functions** and was historically the basis of early lattice signature schemes. You will meet it again properly when we discuss Dilithium/ML-DSA in Session 3 and again in Session 5, since Dilithium's fault-attack surface is closely tied to how it uses a SIS-flavored rejection-sampling step.

## 4 Learning With Errors (LWE)

### 4.1 The Idea, in Plain Language

Imagine someone hands you a stack of linear equations in an unknown secret vector  $s$  — but each equation's right-hand side has been nudged by a small, random amount of noise before you see it. Solving *noise-free* linear equations is easy (that's just linear algebra — Gaussian elimination). Solving them once every equation is slightly wrong, and you don't know by how much, turns out to be extremely hard. That is the entire idea of LWE.

### Remark: Why Not Just Solve It With Linear Algebra?

This is the single most common point of confusion about LWE, so it's worth working through a tiny concrete case. Suppose  $n = 2$  and, with no noise at all, we're given:

$$1 \cdot s_1 + 2 \cdot s_2 = 8, \quad 3 \cdot s_1 + 1 \cdot s_2 = 9.$$

Two equations, two unknowns — solve normally (e.g. elimination) and you get  $s_1 = 2, s_2 = 3$  exactly, every time, no matter how large  $n$  gets, as long as you have  $n$  independent equations. **Linear algebra doesn't care how big the system is** — Gaussian elimination scales smoothly to thousands of unknowns. Now add noise: suppose the equations you're actually given are

$$1 \cdot s_1 + 2 \cdot s_2 = 8 + e_1, \quad 3 \cdot s_1 + 1 \cdot s_2 = 9 + e_2,$$

for some small, unknown  $e_1, e_2 \in \{-1, 0, 1\}$  that you are never told. Now elimination doesn't work: any attempt to “solve exactly” is solving for the *wrong* thing, since the right-hand sides are off by an unknown amount. You could try every combination of  $e_1, e_2$  and re-solve each time — for two small error values that's manageable, but for hundreds of equations, each with its own small unknown error, the number of combinations to check explodes exponentially. That explosion, not any deep mystery, is where LWE's hardness lives: **noise breaks the one tool (linear algebra) that would otherwise make this trivial**, and no comparably efficient tool has been found to replace it.

## 4.2 Formal Definition

### Definition: Learning With Errors (LWE)

Fix a dimension  $n$ , a modulus  $q$ , and an error distribution  $\chi$  over  $\mathbb{Z}_q$  (typically a discrete Gaussian concentrated near 0). Let  $s \in \mathbb{Z}_q^n$  be a fixed secret vector.

- **Search-LWE:** given many samples  $(a_i, b_i)$  where  $a_i \in \mathbb{Z}_q^n$  is uniformly random,  $e_i \leftarrow \chi$  is a fresh small error, and

$$b_i = \langle a_i, s \rangle + e_i \pmod{q},$$

find  $s$ .

- **Decision-LWE:** distinguish samples  $(a_i, b_i)$  generated as above from samples where  $b_i$  is instead *uniformly random* in  $\mathbb{Z}_q$ , independent of  $a_i$ .

Written in matrix form for  $m$  samples stacked together: given  $A \in \mathbb{Z}_q^{m \times n}$  (rows are the  $a_i$ ) and  $b = As + e \pmod{q}$  (with  $e \leftarrow \chi^m$ ), search-LWE asks you to recover  $s$ .

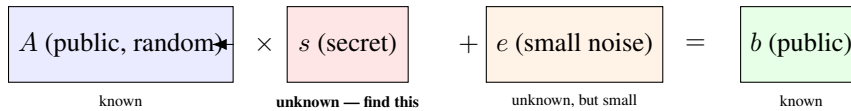


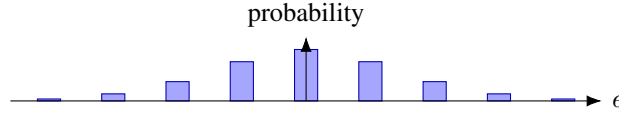
Figure 2: The anatomy of one LWE equation  $b = As + e \pmod{q}$ . An attacker sees  $A$  and  $b$  (both public/known) and must recover  $s$ , without ever being told  $e$  — only that  $e$  is small. Without the noise term  $e$ , this would be ordinary linear algebra and trivial to invert; the noise is precisely what makes it hard.

## 4.3 What the Error Distribution $\chi$ Actually Is

We've been saying “a small error” without pinning down what “small” means or where  $e_i$  actually comes from. Time to fix that.

### Definition: Discrete Gaussian Distribution (the usual choice of $\chi$ )

For a parameter  $\sigma > 0$  (roughly, how spread out the noise is), the discrete Gaussian distribution over  $\mathbb{Z}$  assigns to each integer  $x$  a probability proportional to  $\exp(-x^2/2\sigma^2)$  — the same bell-curve shape as the ordinary (continuous) Gaussian, just evaluated only at integers and re-normalized so the probabilities sum to 1. In practice this means:  $e = 0$  is the single most likely outcome,  $e = \pm 1$  is somewhat less likely,  $e = \pm 2$  less likely still, and values far from 0 (say  $|e| > 6\sigma$ ) are so improbable they essentially never occur.  $\chi$  is this distribution, and each  $e_i$  in an LWE sample is drawn independently from it.



### Remark: Why a Bell Curve, and Not (Say) a Coin Flip $\{-1, 0, 1\}$ ?

Our worked examples below use a crude stand-in ( $e \in \{-1, 0, 1\}$ , each roughly equally likely) purely because it's easy to compute by hand. Real schemes use something bell-curve-shaped (a true discrete Gaussian, or in Kyber's case a closely related and cheaper-to-sample *centered binomial distribution*) for a precise mathematical reason: the hardness proofs connecting LWE back to worst-case lattice problems (Regev's reduction, Section 4.4) require the noise to behave like a Gaussian. Swap in a different-shaped distribution and those proofs may simply no longer apply — “small errors” alone isn't enough; their *shape* matters too. This is a recurring theme you'll meet again in Session 5: real implementations must sample from exactly the right distribution, and subtle sampling mistakes (or side-channel leakage *during* sampling) can quietly undermine security guarantees that look airtight on paper.

## 4.4 A Fully Worked Numeric Example

### Example: Search-LWE by Hand, $n = 3, q = 17$

Let the (unknown, to an attacker) secret be  $s = (2, 9, 1) \in \mathbb{Z}_{17}^3$ , and let the error distribution  $\chi$  produce values in  $\{-1, 0, 1\}$  only (a toy stand-in for a discrete Gaussian). Suppose four samples are generated:

| $a_i$     | $\langle a_i, s \rangle \bmod 17$ | $e_i$ | $b_i = \langle a_i, s \rangle + e_i \bmod 17$ |
|-----------|-----------------------------------|-------|-----------------------------------------------|
| (1, 0, 0) | 2                                 | +0    | 2                                             |
| (0, 1, 0) | 9                                 | -1    | 8                                             |
| (1, 1, 1) | 12                                | +1    | 13                                            |
| (3, 2, 1) | $6 + 18 + 1 = 25 \equiv 8$        | 0     | 8                                             |

An attacker only ever sees the  $a_i$  (left column) and  $b_i$  (right column) — *not* the middle two columns. Notice that sample 1 alone,  $b_1 = 2$ , looks like it might reveal  $s_1 = 2$  directly — and here it does, since  $e_1$  happened to be 0. But sample 2 gives  $b_2 = 8$ , while the true  $s_2 = 9$ : the noise  $e_2 = -1$  has already shifted the visible answer away from the truth by one full step, with no way for an outside observer to know whether the “off-by-one” is coming from  $s_2$  or from  $e_2$ . Multiply that ambiguity across hundreds of dimensions and thousands of samples, and brute-force guessing collapses — this is the essence of why LWE is hard even though each individual equation looks almost solvable.

### Exercise: Try This Before Next Lecture

Using the same  $A$  rows as above but a *different* secret  $s' = (2, 8, 1)$  (differs from  $s$  only in the middle coordinate), recompute  $b_1, \dots, b_4$  with the same errors  $e_i$ . Compare the resulting  $b_i$  values to the table above. How many of the four samples end up identical to the original table, purely by coincidence of the noise? What does this tell you about how many samples an attacker needs before guessing becomes hopeless?

### Remark: Making “Brute Force Collapses” Concrete

In the toy example above, each error  $e_i$  took one of only 3 values, and there were 4 samples. A brute-force attacker who doesn't know the errors could, in principle, try every combination of errors, undo them, and check whether the resulting system solves consistently. The number of combinations to try is  $(\text{number of error values})^{(\text{number of samples})}$ . The table below shows how fast this explodes as the problem scales toward realistic sizes (real schemes use hundreds of samples, and a Gaussian  $\chi$  with far more than 3 realistic values — this table deliberately keeps 3 values fixed just to isolate the effect of *sample count* growing):

| Samples               | Combinations ( $3^{\text{samples}}$ ) | Roughly                          |
|-----------------------|---------------------------------------|----------------------------------|
| 4 (our example)       | 81                                    | trivial by hand                  |
| 20                    | $\approx 3.5$ billion                 | a laptop, seconds                |
| 100                   | $\approx 5 \times 10^{47}$            | far beyond any computer on Earth |
| 500 (realistic scale) | astronomically larger still           | meaningless to even state        |

This is only a crude illustration (real attacks are smarter than raw brute force — that's what lattice-reduction algorithms like LLL and BKZ are for), but it captures the right intuition: **the ambiguity introduced by unknown noise compounds multiplicatively with every additional equation**, which is exactly the opposite of ordinary linear algebra, where more equations make life *easier*, not harder.

### Example: Decision-LWE: Can You Spot the Real Samples?

Here are two sets of  $(a_i, b_i)$  pairs over  $\mathbb{Z}_{17}$ . One set was generated as genuine LWE samples  $b_i = \langle a_i, s \rangle + e_i$  for some fixed secret  $s$ ; the other set has every  $b_i$  replaced by an independent uniformly random value in  $\mathbb{Z}_{17}$ , with no relationship to  $a_i$  at all.

| Set X     |       | Set Y     |       |
|-----------|-------|-----------|-------|
| $a_i$     | $b_i$ | $a_i$     | $b_i$ |
| (1, 0, 0) | 2     | (1, 0, 0) | 11    |
| (0, 1, 0) | 8     | (0, 1, 0) | 3     |
| (1, 1, 1) | 13    | (1, 1, 1) | 6     |

Try to decide, just by looking, which set is “real” LWE and which is uniform noise. (Set X is, in fact, the real one — it's the same data as our search-LWE example above.) **The point of this exercise is that you can't tell, and neither can a computer, efficiently** — both sets look like an unremarkable list of numbers in  $\mathbb{Z}_{17}$ . That indistinguishability, formalized, is decision-LWE. It's also exactly the property real encryption schemes lean on: a ciphertext under LWE-based encryption should be computationally indistinguishable from random noise to anyone without the secret key.



## 4.5 Why LWE Is Believed Hard: the Lattice Connection

### Remark: LWE Is Bounded-Distance Decoding (BDD), a Cousin of CVP

Define the lattice  $\Lambda_q(A) = \{y \in \mathbb{Z}_q^m : y \equiv As \pmod{q} \text{ for some } s \in \mathbb{Z}_q^n\}$  (all vectors reachable as  $A$  times *some* secret, viewed as a lattice via the ring structure of  $\mathbb{Z}_q$  from Lecture 1). The vector  $b = As + e$  is, by construction, a point  $As \in \Lambda_q(A)$  that has been nudged by a *small* vector  $e$ . Recovering  $s$  from  $b$  is therefore exactly **Bounded-Distance Decoding**: given a target  $b$  that is guaranteed to be *unusually close* to some lattice point (closer than a typical random point would be), find that lattice point. This is CVP’s easier, promise-bearing cousin — and, exactly as with CVP in Section 2.2, solving it reliably requires something like a good basis for  $\Lambda_q(A)$ . The whole point of choosing  $A$  uniformly at random is that no such good basis is visible to an attacker; only whoever *generated*  $s$  (or knows a matching trapdoor) has an edge.

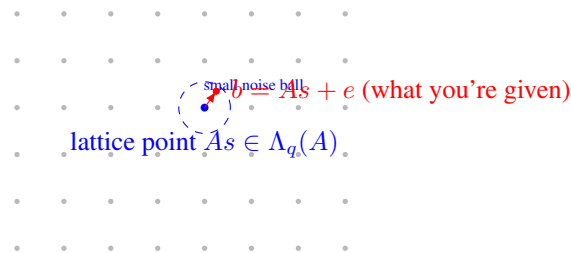


Figure 3: LWE as Bounded-Distance Decoding:  $b$  is guaranteed to land inside a small ball around some lattice point  $As$  — unlike general CVP (Figure 1), where the target could be anywhere. That extra promise (“close, not just closest”) is what makes BDD tractable *with a good basis* and still believed hard *without one*.

### Fact: Regev’s Worst-Case-to-Average-Case Reduction (Statement Only)

A landmark 2005 result (Oded Regev) shows: if there existed an efficient algorithm that solves *average-case, randomly-generated* decision-LWE instances, that algorithm could be turned into an efficient algorithm for *worst-case* GapSVP and other hard lattice problems (via a quantum reduction; later work gave classical reductions for related problems). In plain terms: **LWE’s hardness is not a bet made up from nothing** — it inherits its credibility from the same worst-case lattice hardness Lecture 1 built toward. This is exactly why cryptographers trust random LWE instances, generated fresh for every key, to be hard: hardness on average is tied by proof to hardness in the worst case. We take this as given; the proof itself is well beyond this course.

### Looking Ahead: Search vs. Decision — Also Provably Equivalent

It’s also a known theorem (not proven here) that search-LWE and decision-LWE are polynomially equivalent: an efficient solver for one can be converted into an efficient solver for the other. This matters practically because security *proofs* for encryption schemes are usually stated against decision-LWE (“ciphertexts are indistinguishable from random”), while *attacks* usually aim at search-LWE (“recover the actual secret key”). Session 5 will use both framings, so it’s worth knowing they’re two views of the same underlying hardness.

## 5 From Plain LWE to Ring-LWE

### 5.1 The Problem With Plain LWE: Size

A single LWE public key with security parameter  $n$  requires an  $m \times n$  matrix  $A$  over  $\mathbb{Z}_q$  — roughly  $n^2$  numbers. For cryptographically secure parameters ( $n$  in the many hundreds), that’s an enormous public



key, and every encryption/decryption operation costs a full matrix-vector multiplication,  $O(n^2)$  work. Real deployed schemes need public keys measured in kilobytes and operations fast enough for a TLS handshake or a smartcard, not this.

### Example: Putting Real Numbers On “Too Big”

Take a security-relevant dimension,  $n = 1024$ , with  $q$  a few thousand (so each number needs about 12 bits  $\approx 1.5$  bytes to store). A plain-LWE public key needs roughly  $n^2 = 1,048,576$  such numbers — over 1.5 megabytes, just for one public key, before even discussing ciphertexts. Compare that to a real ML-KEM-1024 public key, which is about 1,568 *bytes* in total — roughly a thousand times smaller. That gap is exactly what Ring-LWE and Module-LWE close, by replacing an  $n \times n$  grid of independent numbers with a much smaller number of *structured* ring elements that each silently represent  $n$  numbers at once (Section 5.3’s remark makes precise how).

## 5.2 The Fix: Move Into a Polynomial Ring

Recall from Lecture 1’s “Ring” section and its preview box: Kyber and Dilithium do arithmetic inside  $R_q = \mathbb{Z}_q[x]/(x^n + 1)$  — polynomials of degree  $< n$  with coefficients in  $\mathbb{Z}_q$ , where powers of  $x$  wrap around using the rule  $x^n \equiv -1$ .

### Definition: The Ring $R_q = \mathbb{Z}_q[x]/(x^n + 1)$

Elements of  $R_q$  are polynomials  $a(x) = a_0 + a_1x + a_2x^2 + \cdots + a_{n-1}x^{n-1}$  with each coefficient  $a_i \in \mathbb{Z}_q$ . Addition is ordinary coefficient-wise addition mod  $q$ . Multiplication is ordinary polynomial multiplication, *followed by reduction*: whenever a term  $x^n$  appears, replace it with  $-1$  (and more generally  $x^{n+k} \rightarrow -x^k$ ), then reduce every coefficient mod  $q$ . This makes  $R_q$  a finite ring with exactly  $q^n$  elements — and, crucially, a single element of  $R_q$  packs  $n$  numbers into one object that a computer can multiply in roughly  $O(n \log n)$  time using the Number-Theoretic Transform (NTT, a finite-field cousin of the FFT) — we will not need NTT details in this course, just the fact that it’s the reason Ring-LWE is fast in practice.

### Example: Multiplying Two Small Polynomials in $R_q$ , $n = 4$ , $q = 17$

Let  $n = 4$ , so we reduce using  $x^4 \equiv -1$ , and work mod  $q = 17$ . Take  $a(x) = 1 + 2x$  and  $c(x) = 3 + x^3$  in  $R_{17}$ . Ordinary polynomial multiplication first:

$$a(x)c(x) = (1 + 2x)(3 + x^3) = 3 + x^3 + 6x + 2x^4.$$

Now reduce using  $x^4 \equiv -1$ : the term  $2x^4$  becomes  $2 \cdot (-1) = -2$ . So we’re left with

$$3 + 6x + x^3 - 2 = 1 + 6x + 0x^2 + x^3.$$

Finally reduce coefficients mod 17 (here every coefficient is already in  $\{0, \dots, 16\}$ , so nothing changes): the product is  $1 + 6x + x^3 \in R_{17}$ . Notice the “wraparound” step ( $x^4 \rightarrow -1$ ) is exactly what folds the high-degree term back down to degree  $< 4$  — this is the polynomial-ring analogue of “carrying” in ordinary mod- $q$  arithmetic.

### 5.3 Ring-LWE, Formally

#### Definition: Ring Learning With Errors (RLWE)

Fix  $R_q = \mathbb{Z}_q[x]/(x^n + 1)$  and an error distribution  $\chi$  over  $R_q$  (coefficients drawn independently from a small distribution). Let  $s \in R_q$  be a fixed secret ring element. A search-RLWE sample is a pair  $(a, b) \in R_q \times R_q$  where  $a$  is uniform random and

$$b = a \cdot s + e \pmod{q}, \quad e \leftarrow \chi.$$

Search-RLWE asks for  $s$  given many such pairs; decision-RLWE asks to distinguish  $(a, b)$  from a uniformly random pair in  $R_q \times R_q$ .

#### Remark: One Ring-LWE Sample $\approx n$ Plain-LWE Samples

Compare the two definitions side by side: plain LWE needs an entire matrix  $A \in \mathbb{Z}_q^{m \times n}$  to state  $m$  equations. A *single* Ring-LWE pair  $(a, b)$ , because  $a$  and  $s$  are each secretly a length- $n$  coefficient vector multiplied together via the structured polynomial product above, is algebraically equivalent to about  $n$  ordinary LWE-style equations bundled into one ring multiplication. This is exactly where the size and speed win comes from — and exactly why the ring's extra structure is a trade-off: it buys efficiency at the cost of a stronger (though still, so far, unbroken) hardness assumption than plain LWE, since the extra algebraic structure is, in principle, more surface area for a future attack to exploit.

## 6 Module-LWE: the Middle Ground Kyber and Dilithium Actually Use

#### Definition: Module Learning With Errors (MLWE)

Fix  $R_q$  as above and a small integer  $k$  (the **module rank**). The secret is now a length- $k$  vector of ring elements,  $\vec{s} \in R_q^k$ . A sample is  $(\vec{a}, b)$  with  $\vec{a} \in R_q^k$  uniform random and

$$b = \langle \vec{a}, \vec{s} \rangle + e \pmod{q} = \sum_{i=1}^k a_i s_i + e, \quad e \leftarrow \chi.$$

#### Remark: MLWE Sits Exactly Between Plain LWE and Ring-LWE

Set  $k = 1$  and MLWE collapses to Ring-LWE exactly. Let  $n = 1$  (no ring structure at all — just plain integers) and MLWE collapses toward plain LWE. Choosing  $k > 1$  with a modest per-ring dimension  $n$  gives designers a dial: more of the speed benefit of rings, without committing to the single largest, most algebraically rigid ring dimension. This is precisely why Kyber and Dilithium use MLWE with a moderate fixed  $n = 256$  and vary  $k$  (Kyber uses  $k \in \{2, 3, 4\}$  for its three security levels) instead of varying  $n$  directly — a design choice we will see in full in Session 3.

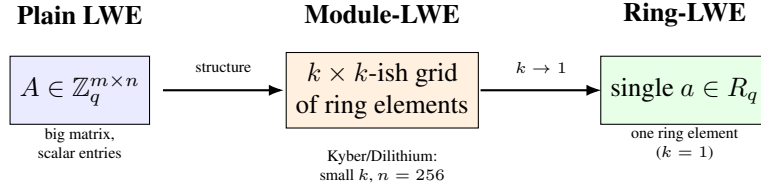


Figure 4: Module-LWE is a size/speed dial between the two extremes: no ring structure at all (plain LWE, left) and maximal ring structure (Ring-LWE, right). Kyber and Dilithium sit in the middle deliberately.

## 7 Bridge to Session 3: A Real Public Key *Is* an (M)LWE Sample

### Looking Ahead: What You'll Recognize Next Session

When we open up Kyber (ML-KEM) in Session 3, its key generation will look almost exactly like the MLWE definition above: pick a random public matrix (of ring elements)  $A$ , pick a small secret  $\vec{s}$  and small error  $\vec{e}$ , and publish  $\mathbf{t} = A\vec{s} + \vec{e}$  as the public key — literally an MLWE sample, with  $\vec{s}$  kept as the secret key. Encryption will use a *second*, independent MLWE-style sample to hide the message inside more noise. Once you can point at a real Kyber public key and say “that’s  $b = As + e$ ,” the scheme stops looking like magic and starts looking like an application of exactly the object we defined today.

## 8 Summary Table

| Problem      | Given                                    | Find / Decide                          |
|--------------|------------------------------------------|----------------------------------------|
| SVP          | Basis $B$ of lattice $L$                 | Shortest nonzero $v \in L$             |
| CVP          | Basis $B$ , target $t \notin L$          | Closest $v \in L$ to $t$               |
| SIS          | Random $A \in \mathbb{Z}_q^{n \times m}$ | Short nonzero $x$ : $Ax \equiv 0$      |
| Search-LWE   | $(A, b = As + e)$                        | Recover secret $s$                     |
| Decision-LWE | $(A, b)$                                 | Is $b$ “ $As + e$ ” or uniform random? |
| Ring-LWE     | $(a, b = as + e) \in R_q^2$              | Recover $s \in R_q$                    |
| Module-LWE   | $(\vec{a}, b) \in R_q^k \times R_q$      | Recover $\vec{s} \in R_q^k$            |

## 9 Full List of Exercises (Assign as Homework Before Lecture 3)

### Exercise: Homework Set

1. State, in your own words (2–3 sentences each), the difference between SVP and CVP, and the difference between the search and decision versions of a problem in general.
2. For  $n = 1$ ,  $m = 3$ ,  $q = 13$ ,  $A = \begin{pmatrix} 4 & 6 & 9 \end{pmatrix}$ : find a nonzero, short  $x \in \mathbb{Z}^3$  solving the SIS instance  $Ax \equiv 0 \pmod{13}$  (Section 3 worked example, above). Confirm your answer by direct computation.
3. Redo the search-LWE worked example (Section 4.3) with secret  $s = (2, 9, 1)$  but a *new* sample  $a_5 = (2, 0, 1)$  and error  $e_5 = 1$ . Compute  $b_5$ . If you were only given  $a_5$  and  $b_5$  (not  $s$  or  $e_5$ ), could you determine  $s$  from this single equation alone? Why not?
4. Explain in your own words why LWE would become *easy* (not hard) if the error  $e$  were removed entirely (i.e.,  $b = As$  exactly). Connect your answer to ordinary linear algebra.
5. In the ring  $R_{17} = \mathbb{Z}_{17}[x]/(x^4 + 1)$ : compute the product  $(2 + x^2)(1 + 3x)$ , showing the wraparound step explicitly (Section 5.2 worked example, above).
6. In 3–4 sentences: why does moving from plain LWE to Ring-LWE reduce key size roughly from  $O(n^2)$  numbers to  $O(n)$  numbers? Use the “one Ring-LWE sample  $\approx n$  plain-LWE samples” remark (Section 5.3) in your explanation.
7. Module-LWE with  $k = 1$  is exactly Ring-LWE, and (informally)  $R_q$  with  $n = 1$  is exactly  $\mathbb{Z}_q$ . Using these two facts, explain why Module-LWE is fairly described as “a generalization that contains both plain LWE and Ring-LWE as special cases.”
8. **Looking ahead:** Kyber-768 uses module rank  $k = 3$  with per-ring dimension  $n = 256$  and  $q = 3329$ . Without looking anything up yet, estimate how many total  $\mathbb{Z}_q$ -coefficients are packed into one Kyber-768 secret key  $\vec{s} \in R_q^3$ . (You’ll check this answer in Session 3.)